Jasper Buntinx

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EDUCATION

Texas A&M University | *Bachelor of Science, Mechanical Engineering, Mathematics*(GPA: 3.31 / 4.0) College Station, TX Aug 2023 - May 2027

SKILLS

- **Software:** SolidWorks, CSWA Certified, AutoCAD, OnShape, SolidWorks Simulation, NX, ANSYS, MATLAB, ParaView, Nvidia Nsight Compute, MS Office Suite, SolidWorks CAM, OrcaSlicer, Fusion 360 CAM, Lightburn, Mastercam
- **Programming:** CUDA, C++, Python, Java, HTML, Arduino/Embedded C
- **AI Tooling:** Claude Code, Hermes, Cursor, API Integration, Multi-Agent Orchestration, Token Optimization
- **Fabrication:** GD&T, FEA, 3D printing, Woodworking, Metalworking, Soldering, MIG Welding, Injection Molding, CNC, Hand tools

WORK EXPERIENCE

Rochester Sensors (Amphenol Subsidiary) — **May 2026 - Present**
Mechanical Engineering Intern — *Dallas, TX*

- Programmed CNC machining workflows in CamWorks for 10+ production parts and dedicated multi-part fixtures, replacing inherited pattern-projected toolpaths to cut cycle times 70–80% (182 → 34 min on highest-volume part) while holding all GD&T requirements.
- Resolved tolerance stack-up and assembly fit defects on in-development products in SolidWorks, validating designs by machining prototypes to released dimensions and correcting a source-design misalignment before mass production.
- Executed environmental qualification testing - accelerated thermal cycling (−40 to +125 °C), salt ingress, and vibration - on sensor assemblies for pre-launch client products, demonstrating reliability beyond 50-year equivalent field life against a 30-year design requirement.
- Programmed Universal Robots arm for vision-guided bin picking, using Keyence 3D scanning to grip randomly oriented parts from bulk bins and sort them at 15–30 parts/min, automating the first station of a phased plan to redeploy operator hours.

ECOFIL — **Aug 2025 - Present**
Cofounder/Mechanical Engineer — *College Station, TX*

- Designed a three-stage filament recycler in SolidWorks, applying GD&T to control mating fits and hold extruded diameter to ±0.10 mm, ensuring consistent standard filament quality.
- Fabricated functional prototypes of the shredder, dehydrator, and extruder through 3D printing (OrcaSlicer), CNC milling (SolidWorks CAM), laser cutting (Lightburn), soldering, and MIG welding, providing working models for performance testing and design iteration
- Secured incubator funding and material donations from Aggies Create, Tyrex, and the TAMU Meloy program, subsequently managing project finances and materials through detailed BOMs to complete filament recycler within budget and resource constraints in Excel.
- Utilize Arduino controllers, high-voltage AC motors, linear Hall-effect sensors, MOSFETs, PTC heating elements, thermistors, stepper motors, fans, and hygrometers to enable real-time monitoring and control of the recycling process.

Petrobras — **Jun 2025 - Aug 2025**
Mechanical Engineering Intern — *Florianopolis, Brazil*

Optimized CUDA-based Lattice Boltzmann (LBM) CFD solvers using AI-assisted kernel profiling in NVIDIA Nsight Compute, resolving memory bottlenecks to push throughput past published open-source LBM benchmarks:

- Achieved 400%+ throughput gain (MLUPS) via FP64→FP32 migration and memory-access restructuring, validated against solver tolerances.
- Authored a Python mesh-generation script for fine-to-coarse grid interfaces, raising the stable peak Reynolds number 33% with no throughput loss.
- Diagnosed uncoalesced global memory access via Nsight Compute workload reports, remapping thread-to-data access for 60% better warp efficiency.
- Cut peak shared memory 35% (56KB→36KB) after identifying redundant data staging and enabling aggressive reuse across thread blocks.

Waterloo Greenway Conservancy (Austin Creek Show) — **May 2022 - Jan 2023**
Mechanical Engineering Intern — *Austin, TX*

- Designed four unique CAD assemblies in SolidWorks from client and artist drawings, enabling production of four large metal dragonfly structures with 14-ft+ wingspans after extensive iteration and prototyping.
- Led fabrication of full-scale dragonfly structures by laser-cutting eight unique wing designs via Fusion 360 CAM, welding 200+ joints, soldering wiring for four LED displays, and producing water-resistant electronics packaging, delivering completed installations ready for the show.
- Achieved 30% reduction in peak current-induced forces on creek structures through FEA with SolidWorks Simulation, ANSYS, and dimensional analysis, enhancing high-wind and flood resilience.

PROJECTS

Autonomous Laundry-Picking Cable Claw (Prototype) — **Dallas, TX May 2026 - Present**

- Derived inverse kinematics for a four-cable ceiling robot, sizing NEMA 17 steppers to 2.6× worst-case tension and validating a 4kg payload.
- Modeled eight parametric 3D-printed parts in SolidWorks, including a five-finger tendon-driven TPU gripper engineered to grasp wide array of fabrics/textures.

Automatic Retractable RFID Bike Lock (Prototype) — **College Station, TX Dec 2025 - Present**

- Developed a retractable RFID locking mechanism (Arduino Nano, RFID reader, solenoid actuation, LED/audio feedback) in a 3D-printed enclosure.

VEX U Competition Robots | *Aggie Robotics (Team WHOOP5)* — **College Station, TX Aug 2023 - May 2024**

- Secured 1st place at 2024 VEX U OVER UNDER State Championship and placement at the World Championship, contributing through AutoCAD schematic redesign and iterative development of an extendable friction launcher to retain, launch, or push Tri-balls as needed.

INVOLVEMENT

One Army Men's Organization | *Special Events Logistics Chair* — **Aug 2025 - Present**

- Led large event construction on $30K+ annual budget, taking builds from design/mockup through BOM and fabrication w/ help from 80+ members for final assembly.
- Work year-round on sponsorship acquisition, obstacle construction, and promotion for Gladiator Dash, a 501(c)(3) 5K mud run raising $180K+ annually.

Freshmen Leaders in Progress (FLIP) | *Service Chair* — **Aug 2023 - May 2025**

- Responsible for the recruitment of 56 freshmen, integration of new members into college life, and coordination of volunteer events with entities including local food pantries, schools, and Habitat for Humanity.